ranging from manual to working with service-level objectives (which we cover in detail in chapter 10).

On a related note, how do you know if something is out of the normal? You need a baseline. You can determine the baseline (i.e., "How does the service or system look when everything is healthy?") manually by observing and noting things or by employing more sophisticated techniques, usually referred to as *anomaly detection* (https://www.reddit.com/r/sre/comments/zxdd2y/anomaly_detection/). With the detection basics out of the way, we now focus on how to deal with the incident after learning about it.

7.1.2 Handling the incident

Now that we're in a position to detect when something goes sideways in our cloud-native app by having health and performance monitoring in place, we move on to the question of how to handle an incident. Let's first discuss the "who" and then move on to the "what" and "how." In general, a human needs to be in the loop, assigned to handle the incident. This is oftentimes done via an on-call mechanism. For a certain time period, ranging from hours to a week or maybe even more, the person on call is notified by the monitoring system and needs to get on top of things as quickly as possible.

> **TIP** There are some offerings, sometimes called *AIOps*, that attempt to optimize the human-in-the-loop experience (e.g., Shoreline, https://www.shoreline.io/).

Nowadays, this notification is usually delivered via an app (and in parallel via email, Slack, MS Teams ping, phone call, etc.) and can come at any time of the day or night. You can probably imagine that if such a notification (or page) reaches you at, say, 2 a.m. and you have, say, 15 minutes to check in, you want to make sure you have a structured approach.

There are several things to consider when talking about on-call roles and models:

- *Traditional vs. on call*—In a traditional setting, where development and operations are strictly separated, the operations (ops) team is on call (technical responder). This can cause issues, since not only is the on-call setting demanding, but also, the person most familiar with a certain service is certainly not the ops person on call. Alternatively, one can establish a model where development and on-call are combined. For example, this is how AWS service teams work. This provides incentives for developers to properly instrument their code, enabling on-call to resolve issues as fast as possible.
- *Engineering and communication*—It goes without saying that to fix a technical issue, you need engineers on call. There is, however, another role that may be separated out: communication (incident manager or commander). On one hand, that can be internal communication (keeping stakeholders up to date), and on the other hand, it can refer to communication of the issue to customers. Someone needs to determine if customers are, indeed, impacted (versus